Dissertation on EV Adoption and Policy Incentives

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1 Abstract

Contents

2 Introduction

Electric vehicles (EVs) have emerged as a pivotal technology in the global transition towards sustainable transportation. Governments worldwide have implemented a range of incentives to accelerate EV uptake, such as direct purchase subsidies, tax rebates, and exemptions from registration fees. In addition, indirect incentives such as investments in charging infrastructure, preferential access to urban zones, and public awareness campaigns have played a significant role in shaping consumer behavior. As EV technology continues to evolve, policy incentives remain a critical lever for driving adoption and achieving environmental objectives. As one of the greatest carbon emitters, China committed to carbon neutrality by 2030 and has put significant policy power into environmental policies. China remains the world's largest automobile market, the transportation sector, as a traditional carbon-intensive industry, presents as one of the greatest challenges yet a great opportunity to decarbonization. China's average annual growth rate hits approximately 15% in automobile base since 1990, the momentum in the traditional automobile industry makes the transition toward sustainable transport especially difficult. Therefore, the question of how to efficiently incentivize EV adoption closely links to environmental policy frameworks, which aim to mitigate greenhouse gas emissions and reduce urban air pollution.

2.1 Research Question

This paper aims to answer the following research question:To what extent does the presence of indirect network effects from charging stations influence the substitutional patterns and the effectiveness of policy incentives on EV adoption in China?

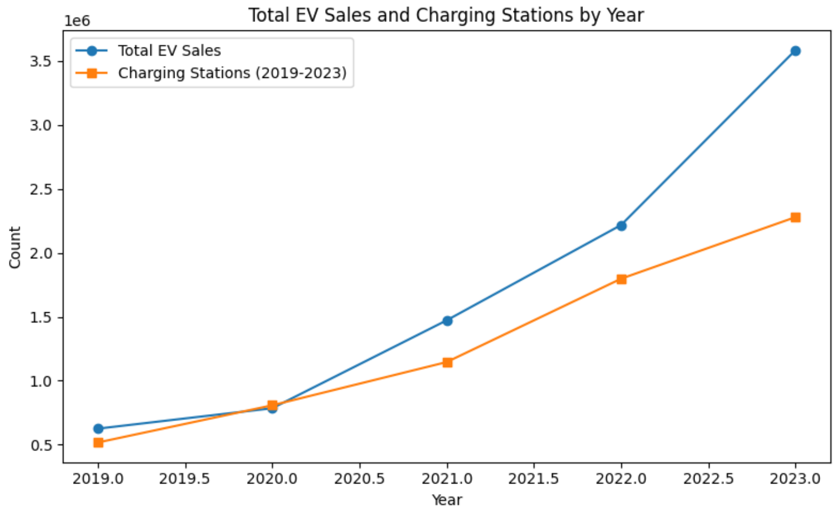

Figure 1: EV Stock and Charging Station Stock in China (2019-2023)

3 literature Review

This study primarily relates to the following three strands of literature.

Differentiated Market Demand estimation

First, this study closely relates to the mature literature on differentiated product demand estimation, in which the automobile market has been specifically focused on with a variety of directions for extensions. Following the foundational work of MacFadden (1977), the strucutral approach of logit discrete choice has been widely adopted for the demand-supply style problems. To allow for a more realistic assumtions and promptly deal with dimensionality problems, variants in the model such as the nested logit model, the principles-of-differentiation general extreme-value (PD-GEV) model in Bresnahan et al. (1997), and the adopted approach in this paper of random-coefficient discrete choice model in Berry et al. (1995). Following the seminal works of Bresnahan (1987), Berry et al. (1995), and Petrin (2002), I apply the standard structural modelling techniques to the automobile industry, in order to recover flexible substitution patterns and facilitate counterfactual policy evaluations. Although automobile demand estimation has been thoroughly discussed in the past three decades, the focus on EV is relatively fresh within the literature. Li et al. (2017) began with a stylized model to evaluate the effectiveness of EV and infrastructure incentives in the U.S. market. They found that charging station subsidies of equal sizes are twice as effective for EV adoption that purchase incentives. Springel (2021) aimed at similar question in the Norwesian market, but implemented a structural demand-supply specification and estimated a random coefficient model under the influence of indirect network externality of charging stations, which then faciliates the testing of counterfactual policy mixes. Li (2023) considered again the U.S. automobile market and EV adoption using a BLP-style model, but introduced incompatible charging standards to the supply side and focused primarily on the impact of a uniform-standard policy for EV penetration. Remmy (2023) expanded the strucutral model by endogenizing supply-side decision on range of EVs and explored firms', in addition to consumers', responses to incentive schemes in Germany. Finally, Heid et al. (2024) enriched the modelling further by bringing in electricity market to form a joint-equilibrium model, endogenizing charging timing on the demand side, and further discussed the issue of pass-through in incentives. As for studies on the Chinese automobile industry, Jia et al. (2023) studied the impact of attribute-based incentives to EV adoption building on a structural demand model. They found that But they focus primarily on the strand of endogenous product attributes and abstracted away from the supply-side decision on charging station entry. To the best of my knowledge, there is yet a systematic study of the Chinese EV market that incorporates indirect network effects.

Indirect Network Effect

The interaction between EV and charging station presents a typical example of a two-sided market, in which the presence of indirect network effect would generally alter policy implications significantly. The literature on indirect network effect traces back to the earlier theoretical works such as Katz and Shapiro (1985) and Farrel and Saloner (1985). This study specifically adds to the strand of automobile demand modelling by testing the result of indirect network effect in the Chinese market.

EV Adoption and Environmental Policy

Lastly, this paper contributes to the understanding of policy impacts on EV adoption. The Chinese government has prompted EV penetration of the largest in scale and fastest in speed through strong policy supports. Although the growing literature of EV adoption and environmental policy has studied cases of various developed economies such as Norway (Springel 2021), the US (Li et al. 2017) (Li 2023), and Germany (Remmy 2024), there is yet to be a systematic study of the evloving Chinese EV market to my knowledge. EV policies have been pivotal to the net-zero target of China, where a wide-range of policies have been entering, substituting and existing in a frequent basis, allowing for many additional policy questions to be answered by evaluating the rich set of policy mix. In particular, the question on policy exit effect is rarely asked for this relatively yound industry, and this paper adds to that direction, by extending the question of optimal policy mix in Springel (2021) to exploring the compensatory effects of policy substitutions.

4 Industrial Background and Policy

4.1 Development of Chinese EV market

The Chinese EV market has experienced a rapid expansion period over the last decade, and various incentives in the form of both central and disaggregated policies have contributed to the process. Back in 2009, the Chinese government first initiated the experimental policy initiative of "1,000 EVs in 10 cities", aiming at introducing new energy vehicles at 10 cities per year. To initialize EV adoption, the replacecment over traditional ICE cars initially took place in public transports, taxi, and other government-led fields. However, EV penetration was unsuccessful during ealier stages due to immature industry of EV production, technologies and infrastures. Within the same period, monetary incentives were also introduced on both supply and demand side in the form of cost rebates and purchase subsidies. In 2014, a uniform tax rebate policy was introduced for all purchases on BEVs and PHEVs. These conitnued to exist until the environmental policies gradually gained public attention in major cities. Many local environmental policies had taken the form of travel and registration restrictions on traditional vehicles while leaving EV unaffected. As technologies progressed, prices effects combining with usage value rise accelerated EV penetration.

In the starting period, EV share rised explosively but with a strong reliance on intense monetary incentives, as a result, subsidy frauds also came as a side effect. As a result, the attention gradually shifted from direct to indirect policies, among which infrastructure supply was a key target due to the unique two-sided market feature of EV. From 2017, policy exists on direct subsides was complemented by construction cost rebate and operational subsidies for charging stations. In the meantime, foreign capital restrictions were lifted for automobile producers, following the entry of Tesla's super-factory in Shanghai, number of brands began to rise, pushing up the intensiveness of differentiated product competition. In addition, the 'Dual Credit Policy' published in 2017 effectively ensured a minimum share for EV from the supply side, resulted in large scale entires of traditional ICE automobile players to expand into EV production.

4.2 Central/local government incentives: purchase and infrastructure

5 Data

The data I use is combiled from a variety of sources, including a commercial source that aggregated registry data from the Chinese Association of Automobile Manufacturers (CAAM), indiative price and range data from the Minstry of Industry and Information Technology (MIIT), charging station stock data from the China Electric Vehicel Charging Infrastcture Promotion Alliance (EVCIPA), the province-level demographic data National Bureau of Statistics (NBS), and supplementary information on a wide-range of government incentives at different levels.

The primary data set is the sales data where each entry contains the number of sales for a particular model within a province in a given year in the range of 2019-2023, it also contains brand, fuel-type and two supplementary product characteristics of each model: mass and power. This data set is over all types of private cars, consisting ICE cars and EVs that can be further divided into BEVs and PHEVs. The data set was purchased from a commercial source that aggregated micro data of Compulsory Traffic Accident Liability Insurance (CTALI) maintained by the CAAM. It is common practice to use this measure as proxy for car purchases in the Chinese market, as CTALI registed cars are the ones actually in-use. The number of models appears to vary across years and the availability of some are not uniform across provinces. I define a market as the unit of observation at the model-year-province level, the data set consisting of ?? models, 31 provinces over a 5-years interval, gives me a total number of ?? observations. This sample size is significant compares to other studies due to the large geographic dimension and uniquely rich diversification in the Chinese automobile market, supported by its market base.

The second data set collected information on indicative price range published by the MITT, and I use the midpoint of which as the market price. It also contains the range of each model.

I also obtain the charging station stock at the province-year level from 2018-2023.

Lastly, I combile mean and standard deviation of demographic variables for each province-year, and use specifically the number of households as a proxy for the market size.

6 Empirical Modelling

6.1 Demand: Vehicle Choice

For modelling vehicle demand side, I follow the tradtitional setup of differentiated product demand estimation to use a random utility maximization setup for modelling of discrete choices. In particular, assume there re $m = 1,...,M$ markets that is defined as a province-year pair within my context of study. Within each of the defined market, the modelling is on individual purchase decisions of $i = 1,...,I_m$ number of potential consumers, made over $j = 1,...,J$ different vehicle models in the product space. The conditional indirect utility of consumer i, $U(x_{jt}, \xi_{jt}, p_{jt}, \tau_i; \theta)$ of consuming product j in market m is specified as a function of both observed and unobserved product characteristics:

$$u_{ijm} = \beta^N \log(N_j m) + \alpha p_{jm} + x_{jm}\beta^{ex} + \xi_{jm} + \bar{\epsilon}_{ijm} \tag{1}$$

$i = 1, ..., I_m,\ j = 0, .., J,\ m = 1, ..., M,$

where p_{jm} denotes the net product price of product j in market m, y_i is the income of consumer, x_{jt} is a K-dimensional row vector of observable produt characteristics of product j, ξ_{jm} is the unobservable characteristics, and $\bar{\epsilon}_{ijm}$ is a mean-zero stochastic term representing the remaining individual-specific valuation for product j. Additionally, an outside good representing no purchases is defined to have indirect utlity of $u_{i0m} = \bar{\epsilon}_{ijm}$. Consumer i chooses between the outside good 0 and other J differentiated products in the market and purchases one unit of good j if and only if $u_{ijm} > u_{ikm}$ for all $k \geq 0$ and $k \neq j$. As a standard practice notation-wise, I define the mean utility level of product j in market m as:

$$\delta_{jm} \equiv \beta^N \log(N_j m) + \alpha p_{jm} + x_{jm}\beta + \xi_{jm} \tag{2}$$

Following the distributional assumptions of the nested logit model, the individual valuation term $\bar{\epsilon}_{ijm}$ can be further decomposed as:

$$\bar{\epsilon}_{ijm} = \zeta_{igm} + (1-\rho)\epsilon_{ijm} \tag{3}$$

Where ζ_{igm} is a group-common term of group $g = 0, ..., G$, and an i.i.d individual valuation shock $(1-\rho)\epsilon_{ijm}$. Here, $0 \leq \rho \leq 1$ is the nesting parameter approximates for the degree of preference correlation between products of the same group. Following the interpretation of Berry (1994), the group-common term ζ_{igm} can be seen as random coefficeints on a set of group-specific dummies d_{jgm} that equals to 1 if j belongs to group g. Summarizing above, consumer i's conditional indrect utility can now be written as:

$$u_{ijm} = \delta_{jm} + \sum_g d_{jgm}\zeta_{jgm} + (1-\rho)\epsilon_{ijm} \tag{4}$$

The nested logit model assumes that the i.i.d shock ϵ_{ijm} follows a type I extreme value distribution, and as shown by Cardell (1997), there exist a unique distribution of ζ_{ig} that $[\zeta_{ig} + (1-\rho)\epsilon]$ also follows an extreme value distribution. The distributional assumptions allow for closed-form solutions for the choice probabilities. In the spirit of Berry (1994), I write the well-known result of choice probability of choosing product j within group g, which equivalent gives the within-group market of product j:

$$s_{j/g} = \frac{e^{\delta_{jm}/(1-\rho)}}{D_g} \tag{5}$$

where D_g is the denominator that normalizes the choice probability, defined as:

$$D_g \equiv \sum_{j \in J_g} e^{\delta_{jm}/(1-\rho)} \tag{6}$$

The probability of choosing group g and hence the market share of group g can be expressed as:

$$s_g = \frac{D_g^{(1-\rho)}}{\sum_g D_g^{(1-\rho)}} \tag{7}$$

Thus, the overall market share of product j in market m reads:

$$s_{jm} = s_{j/g}s_g = \frac{e^{\delta_{jm}/(1-\rho)}}{D_g^{\rho}[\sum_g D_g^{(1-\rho)}]} \tag{8}$$

Specifically, the outside good as the only product in group 0, with normalizations $\delta_0 \equiv 0$ and $D_0 = 1$, has a market share of:

$$s_{0m} = \frac{1}{\sum_g D_g^{(1-\rho)}} \tag{9}$$

From setting out the model and predicted market share, taking natural logarithms of both sides gives an analyitic linear expression for estimation:

$$\log(s_{jm}) - \log(s_{0m}) = \delta_{jm}/(1-\rho) - \delta \log(D_g) \tag{10}$$

Recalling the definition of D_g and δ_{jm} gives the final form:

$$\log(s_{jm}) - \log(s_{0m}) - \rho \log(s_{j/g}) = \beta^N \log(N_j m) + \alpha p_j + x_j \beta + \xi_{jm} \tag{11}$$

The additional term $\rho \log(s_{j/g})$ accounts for the correlation of preferences within groups, which is put to the left of equation indicating its endogeneity as in Conlon and Gortmaker (2020).

6.2 Supply: Station Entry

My specification closely follows that of Springel (2021) and Remmy (2023). Let $h = 1, ..., N_m$ denote the set of charging station firms, and $j = 1, ..., J$ denote the set of vehicle models. Assume firms are price-setters and that each firm maximizes profits over a subset J_f. Assuming a standard profit function that is quasi-concave in price, the per-consumer profit function is given as $D_{hm}(p_{hm}, p_{-hm}, N_m)(p_{hm} - MC_{hm})$, where the per-consumer demand D_{hm} faced by firm f is given by a function of charging station network N_m, own price p_{hm} and the price of competing firms p_{-hm}. Similarly to Springle (2021) and Remmy (2023), I make the following simlifying assumptions in the spirit of Gandal, kende, and Rob (2000) and Bresnahan and Reiss (1991): 1. Symmetric per-consumer demand function; 2. Constant marginal cost MC_{hm} and sunk cost of entry F_{hm} across all firms f in market m; 3. Each station earns and equal share of market profit from symmetry. These assumptions ensures the eixstence of an equilibrium that all stations charge the same price, and the post-entry station profit per-period reads:

$$\pi_m = Q_m^{EV} \frac{D(p(N_m))\phi(N_m)}{N_m} \tag{12}$$

where Q_m^{EV} is the cumulative EV stock market m, $\phi(N_m) \equiv p - MC$ is the equilibrium markup, and $p(N_m)$ is assumed to decrease in N_m. For a simpler notation, denote $f(N_m) \equiv \frac{D(p(N_m))\phi(N_m)}{N_m}$

Next, I assume that the entry decision of charging station firms is made at the beginning of each period and utilizes a free-entry condition to derive the equilirbium relationship. I also modify notation to replace market m with a province-year pair pt for clarity in timing. If a firm enters the market in period t, it incurs a sunk cost of entry F_{pt}, and earns a stream of per-period profits starting next period $(\pi_{pt+1}, \pi_{pt+2}, ...)$. The free-entry condition requires that the expected present value of profits from entering the market is equalized across periods, that is:

$$-F_{pt} + \sum_{s=1}^{\infty} \frac{\pi_{pt+s}}{(1+r)^s} = -\frac{F_{pt+1}}{(1+r)} + \sum_{s=2}^{\infty} \frac{\pi_{pt+s}}{(1+r)^s} \tag{13}$$

Combining the station profit expression with the free-entry condition and taking natural logarithm of both sides, the expression boils down to:

$$\log(f(N_{pt})) = -\log(\frac{1}{(1+r)}) - \log Q_{ct}^{EV} + \log(F_{pt} - \frac{1}{1+r} F_{pt+1}) \tag{14}$$

Additionally, I leverage the unique feature of the policy structure in China, where province governments perform independent subsidizing polilicies of different forms across the observed period, bringing in additional variations to explore policy effectiveness. By specifying $f(N_{pt}) = (bN_{pt})^{-d}$ and assuming the entry cost F_{pt} is a linear function of exogenous cost shifter, province demographics which captured by fixed effect ρ_p, and a time trend $\tau(t)$, the station entry model is specified as:

$$\log(N_{pt}) = \lambda_0 + \lambda_1 \log(Q_{pt}^{EV}) + \lambda_2 FX_{it} + \lambda_3 OP_{it} + \lambda_4 \rho_p + \lambda_5 \tau(t) + \epsilon_{pt} \tag{15}$$

7 Empiricals

7.1 Identification

IV strategies

7.2 Estimation

8 Results

8.1 Demand

Elasticities (absent of indirect network effect)

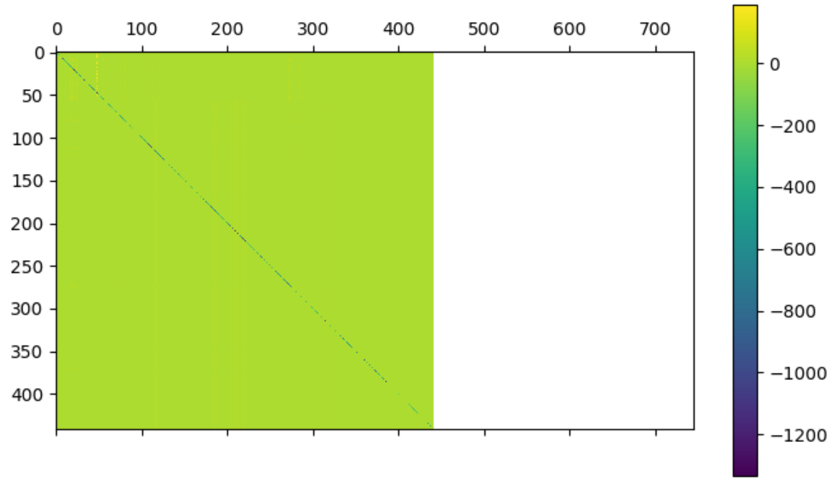

Figure 2: Example of Elasticities (Shanghai, 2019)

Table 1: Rho and Beta Estimates (Robust SEs in Parentheses)

	Rho (All Groups)	prices	is_electric	is_electric*log_charging_IV
Estimate	+9.9E−01	−2.4E−01	−8.6E+00	+6.7E−01
(SE)	(+1.5E−02)	(+2.3E−03)	(+2.2E−01)	(+2.2E−02)

	(1)	(2)
Intercept		11.132***
		(0.382)
$log(EV_stock)$	0.301	
	(1.036)	
time_trend	0.286	
	(0.453)	
shift_share_IV		0.000***
		(0.000)
Fixed Effects	Province-Year	
Observations	155	155
R^2	0.973	0.126
Residual Std. Error	1.747 (df=122)	0.558 (df=153)
F Statistic	1004081.737*** (df=33; 122)	13.667*** (df=2; 153)
Note:		*p<0.1; **p<0.05; ***p<0.01

8.2 Supply

9 Policy Counterfactuals

10 Conclusion